STETHOSCOPE STUDY

HESI A2 & TEAS 7 Master Study Guide

200	200	8	4
HESI A2 Questions	TEAS 7 Questions	HESI Sections	TEAS Sections

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WHAT'S INSIDE

HESI A2 — MATH (25 Questions on Exam)

ESSENTIAL FORMULAS

FORMULA	EQUATION	EXAMPLE
IV Flow Rate (gtt/min)	(Volume mL x Drop factor) / Time in minutes	(500 x 15) / 120 = 62.5 ~ 63 gtt/min
Hourly IV Rate	Total volume / Total hours	1000 mL / 8 hrs = 125 mL/hr
Dose Calculation	(Desired / Available) x Volume	(500/250) x 5 mL = 10 mL
Weight-based Dose	Dose per kg x Patient weight (kg)	0.1 mg/kg x 70 kg = 7 mg
lbs to kg	Pounds / 2.2	176 lbs / 2.2 = 80 kg
kg to lbs	Kilograms x 2.2	80 kg x 2.2 = 176 lbs
F to C	(F - 32) x 5/9	(101.3 - 32) x 5/9 = 38.5 C
C to F	(C x 9/5) + 32	(38.5 x 9/5) + 32 = 101.3 F
Percent	Part / Whole x 100	35 / 50 x 100 = 70%
24-hr Dose	Single dose x Doses per day	650 mg x 4 doses = 2,600 mg/day

CRITICAL CONVERSIONS

METRIC	HOUSEHOLD/CLINICAL	NOTES
1 kg = 1,000 g	1 lb = 16 oz	Always convert lbs to kg before dosing
1 g = 1,000 mg	1 oz = 30 mL	Normal urine output: at least 30 mL/hr
1 mg = 1,000 mcg	1 cup = 240 mL	1 L = 1,000 mL
2.2 lbs = 1 kg	1 tbsp = 15 mL	1 tsp = 5 mL
1 L = 1,000 mL	1 grain = 60 mg	1 inch = 2.54 cm

Nurse Tip: Always convert units FIRST before calculating. If the order is 0.5 g and tablets are 250 mg — convert 0.5 g to 500 mg, then divide. Most math errors happen from skipping the unit conversion step.

MILITARY TIME — Quick Reference

STANDARD	MILITARY	STANDARD	MILITARY
12:00 AM (midnight)	0000	12:00 PM (noon)	1200
1:00 AM	0100	1:00 PM	1300
6:00 AM	0600	6:00 PM	1800
8:00 AM	0800	8:00 PM	2000
11:00 AM	1100	11:00 PM	2300

Rule: 0001-1259 = AM (as-is). 1300-2359 = subtract 1200 to get PM time.

HESI A2 — READING & VOCABULARY (55 Questions Combined)

HIGH-YIELD MEDICAL PREFIXES

PREFIX	MEANING	NURSING EXAMPLES
a-, an-	without, absent	Apnea (no breathing), Afebrile (no fever), Anuria (no urine)
brady-	slow	Bradycardia (HR below 60), Bradypnea (RR below 12/min)
tachy-	fast	Tachycardia (HR above 100), Tachypnea (RR above 20/min)
hypo-	below normal	Hypoglycemia, Hypotension, Hypothermia, Hyponatremia
hyper-	above normal	Hyperglycemia, Hypertension, Hyperthermia, Hyperkalemia
dys-	difficult, painful	Dyspnea, Dysphagia, Dysuria, Dysrhythmia
pre-	before	Preoperative, Prenatal, Precaution
post-	after	Postoperative, Postpartum, Post-ictal
anti-	against	Antibiotic, Antiemetic, Antihypertensive, Antifungal
peri-	around	Pericardium, Perioperative, Peripheral
sub-	under, below	Subcutaneous, Sublingual, Subacute, Subdural
bi-	two, both	Bilateral, Bicuspid (mitral valve), Bimanual
inter-	between	Intercostal, Interdisciplinary, Interpersonal
intra-	within	Intravascular, Intramuscular, Intracranial, Intravenous

HIGH-YIELD SUFFIXES

SUFFIX	MEANING	EXAMPLES
-itis	inflammation	Appendicitis, Gastritis, Arthritis, Cellulitis, Tonsillitis
-ectomy	surgical removal	Appendectomy, Mastectomy, Tonsillectomy, Cholecystectomy
-otomy	surgical incision	Tracheotomy, Laparotomy, Craniotomy
-ostomy	surgical opening	Colostomy, Tracheostomy, Ileostomy
-ology	study of	Cardiology, Neurology, Oncology, Nephrology, Hematology
-pnea	breathing	Apnea, Dyspnea, Tachypnea, Bradypnea, Orthopnea
-uria	urine condition	Hematuria, Oliguria, Polyuria, Dysuria, Glycosuria
-phagia	eating/swallowing	Dysphagia (difficulty), Polyphagia (excess), Aphagia
-algia	pain	Neuralgia, Fibromyalgia, Arthralgia, Myalgia
-emia	blood condition	Hyperglycemia, Anemia, Septicemia, Leukemia

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TRANSITION WORDS — Know What Each Signals

RELATIONSHIP	SIGNAL WORDS
Contrast	however, although, despite, yet, nevertheless, on the other hand, but, conversely, whereas
Cause and Effect	therefore, thus, consequently, as a result, because, since, so, due to
Addition	furthermore, moreover, in addition, also, additionally, besides, and
Example	for example, for instance, such as, to illustrate, specifically, including
Summary	in conclusion, in summary, therefore, overall, in short, to summarize, finally
Sequence/Time	first, then, next, finally, subsequently, after, before, during, meanwhile

HESI A2 — GRAMMAR (25 Questions on Exam)

MOST TESTED GRAMMAR RULES

RULE	CORRECT	INCORRECT
Compound subject + verb	The nurse and doctor were present	The nurse and doctor was present
Object pronoun after preposition	Between you and me	Between you and I
Singular indefinite pronoun	Each nurse is responsible	Each nurse are responsible
Neither/nor — nearest subject	Neither the nurses nor the doctor is available	Neither the nurses nor the doctor are available
Plural possessive apostrophe	The patients' rights (multiple patients)	The patients's rights
Singular possessive apostrophe	The nurse's station	The nurses station
Dangling modifier	Walking down the hall, the nurse heard the alarm	Walking down the hall, the alarm was heard
Parallel structure	assessing, documenting, administering	to assess, documenting, administer
Comma + coordinating conjunction	The patient was stable, and the nurse documented	The patient was stable and, the nurse documented
Semicolon — two related clauses	The patient refused; the nurse documented it	The patient refused, however the nurse documented it

FANBOYS = Coordinating Conjunctions: For • And • Nor • But • Or • Yet • So Use a comma BEFORE a FANBOYS word only when joining two complete independent clauses. Active voice is preferred in clinical documentation: "The nurse administered" not "The medication was administered."

HESI A2 — BIOLOGY & CHEMISTRY (40 Questions Combined)

CELL ORGANELLES — Know Every Function

ORGANELLE	FUNCTION	MEMORY TRICK
Nucleus	Control center; contains DNA; directs cell activities	"Brain" of the cell
Mitochondria	Produces ATP (energy) via cellular respiration	Powerhouse of the cell
Ribosome	Protein synthesis — translates mRNA into proteins	Ribosome = pRoteins
Rough ER	Processes and transports proteins (has ribosomes attached)	Rough = ribosomes on it
Smooth ER	Lipid synthesis and detoxification (no ribosomes)	Smooth = no ribosomes
Golgi Apparatus	Packages, modifies and ships proteins to destinations	Post office of the cell
Lysosome	Digests cellular waste, old organelles, and pathogens	Garbage disposal
Cell membrane	Selectively permeable barrier — controls what enters/exits	Gatekeeper
Vacuole	Stores water, nutrients, and waste products	Storage unit
Chloroplast	Photosynthesis — converts sunlight to glucose (plants only)	Has chlorophyll (green)

TRANSPORT MECHANISMS

TYPE	ATP?	DIRECTION	EXAMPLE
Simple diffusion	No — passive	High to low concentration	O2 into cells, CO2 out
Osmosis	No — passive	Water moves from low solute to high solute	Water across cell membrane
Facilitated diffusion	No — passive	High to low via protein channel	Glucose into cells (GLUT transporters)
Active transport	YES — needs ATP	Low to high (against gradient)	Sodium-potassium pump in neurons
Endocytosis	YES	Into cell (engulfs material)	Phagocytosis of bacteria by WBCs
Exocytosis	YES	Out of cell (secretes material)	Neurotransmitter release at synapses

CHEMISTRY FUNDAMENTALS

CONCEPT	KEY FACTS	NURSING RELEVANCE
pH Scale	pH 0-6 = acidic. pH 7 = neutral. pH 8-14 = basic/alkaline.	Blood pH 7.35-7.45 — know for ABG interpretation
Acids	Donate H+ ions. pH below 7. Taste sour.	HCl in stomach. Acidosis on ABGs (pH below 7.35)

CONCEPT	KEY FACTS	NURSING RELEVANCE
Bases	Accept H+ ions. pH above 7. Taste bitter.	$NaHCO_3$ (bicarb). Alkalosis on ABGs (pH above 7.45)
Buffer Systems	Resist changes in pH by absorbing excess H+ or OH-.	Bicarbonate buffer maintains blood pH 7.35-7.45
Ionic Bond	One atom TRANSFERS electrons to another. Creates ions.	NaCl (table salt): Na+ and Cl- — electrolytes in blood
Covalent Bond	Atoms SHARE electron pairs. Stronger than ionic.	H_2O, organic molecules, DNA backbone
Hydrogen Bond	Weak attraction between polar molecules.	Holds DNA double helix strands together
Enzyme	Biological catalyst. Speeds up reactions without being consumed.	Catalase breaks down H_2O_2. Inhibited by extreme temp or pH
Neutralization	Acid + Base yields Salt + Water.	HCl + NaOH yields NaCl + H_2O. Basis of body buffer systems

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HESI A2 — ANATOMY & PHYSIOLOGY (30 Questions on Exam)

CARDIOVASCULAR SYSTEM

STRUCTURE / CONCEPT	KEY FACTS
SA Node	Natural pacemaker in right atrium. Fires 60-100 bpm. Initiates every heartbeat.
AV Node	Delays impulse ~0.1 sec to allow ventricles to fill before contracting.
Right heart blood flow	Body > Vena Cava > Right Atrium > Tricuspid valve > Right Ventricle > Pulmonic valve > Pulmonary Artery > Lungs
Left heart blood flow	Lungs > Pulmonary Veins (4) > Left Atrium > Mitral valve > Left Ventricle > Aortic valve > Aorta > Body
Cardiac Output (CO)	CO = Heart Rate x Stroke Volume. Normal approximately 5 L/min at rest.
Normal Blood Pressure	Normal below 120/80. Hypertension at or above 130/80. Hypertensive crisis above 180/120.
Normal Heart Rate	60-100 bpm. Bradycardia below 60. Tachycardia above 100. Both require assessment.
Right-side valves	Tricuspid (right atrium to right ventricle). Pulmonic (right ventricle to pulmonary artery).
Left-side valves	Mitral/Bicuspid (left atrium to left ventricle). Aortic (left ventricle to aorta).

RESPIRATORY SYSTEM

CONCEPT	KEY FACTS
Gas exchange site	Alveoli — tiny air sacs at end of respiratory tree. O_2 diffuses in, CO_2 diffuses out. Surfactant prevents collapse.
Normal RR (adult)	12-20 breaths/min. Tachypnea above 20. Bradypnea below 12. Below 8 = respiratory emergency — act immediately.
Normal O_2 saturation	95% or above. Concern below 92%. Critical below 88% — apply supplemental oxygen immediately.
Epiglottis	Cartilage flap that covers the larynx during swallowing. Prevents food/liquid aspiration into trachea.
Carina	Ridge where trachea splits into left and right primary bronchi (at T4 level on chest X-ray).
Surfactant	Produced by Type II pneumocytes. Reduces surface tension in alveoli. Premature infants lack it causing RDS.
Control of breathing	Medulla oblongata (brainstem) — primarily responds to rising CO_2 levels in blood.

ENDOCRINE SYSTEM — Hormones to Know

GLAND	HORMONE	FUNCTION	IMBALANCE
Anterior Pituitary	TSH, ACTH, GH, FSH, LH	Stimulates other endocrine glands	Pituitary tumor can cause acromegaly (excess GH)
Posterior Pituitary	ADH (vasopressin), Oxytocin	Water reabsorption; uterine contractions in labor	SIADH (excess ADH) causes hyponatremia from water retention

GLAND	HORMONE	FUNCTION	IMBALANCE
Thyroid	T3 and T4 (require iodine)	Regulates metabolism, energy, growth and development	Hypothyroid = slowed metabolism. Hyperthyroid = accelerated metabolism.
Parathyroid	PTH (parathyroid hormone)	Raises blood calcium level	Hypoparathyroid causes hypocalcemia and tetany (muscle spasms)
Adrenal Cortex	Cortisol, Aldosterone	Stress response; sodium retention; blood pressure	Cushing's (excess cortisol). Addison's disease (deficiency).
Adrenal Medulla	Epinephrine, Norepinephrine	Fight-or-flight response	Pheochromocytoma causes hypertensive crisis
Pancreas (beta)	Insulin	Lowers blood glucose — facilitates cell uptake	Type 1 DM (no insulin). Type 2 DM (resistance to insulin).
Pancreas (alpha)	Glucagon	Raises blood glucose — triggers glycogen breakdown	Critical role in preventing hypoglycemia between meals

A&P; CONTINUED — Nervous, Renal and Musculoskeletal Systems

NERVOUS SYSTEM

CONCEPT	KEY FACTS
Brain lobes	Frontal (movement, personality, executive function). Parietal (sensory processing). Temporal (hearing, memory). Occipital (vision).
Brainstem	Medulla oblongata = vital functions (HR, RR, BP). Pons = sleep, bladder. Midbrain = eye movement, pain relay.
Cerebellum	Balance, coordination, and fine motor control. Damage causes ataxia (uncoordinated movement).
Sympathetic NS (fight-or-flight)	Increases HR and BP. Dilates pupils and bronchioles. Redirects blood to muscles. Inhibits digestion.
Parasympathetic NS (rest-and-digest)	Decreases HR. Promotes digestion. Constricts pupils. SLUDD: Salivation, Lacrimation, Urination, Defecation, Digestion.
Reflex arc	Sensory neuron > spinal cord interneuron > motor neuron. Bypasses brain for speed. Example: patellar (knee-jerk) reflex.
Vagus nerve (CN X)	Parasympathetic control of heart (slows HR), lungs, and most abdominal organs. Most widespread cranial nerve.

RENAL SYSTEM

CONCEPT	KEY FACTS
Nephron	Functional unit of kidney. Approximately 1 million per kidney. Filters, reabsorbs, and secretes.
Nephron pathway	Glomerulus > Bowman capsule > Proximal tubule > Loop of Henle > Distal tubule > Collecting duct > Renal pelvis > Ureter
ADH effect	Increases water reabsorption in collecting duct. Results in decreased urine output and concentrated urine.
Aldosterone effect	Increases sodium reabsorption (water follows sodium). Increases blood volume and blood pressure.
Normal urine output	30 mL/hr or more in adults. Oliguria = below 30 mL/hr. Anuria = no urine. Both require immediate reporting.
RAAS system	Kidneys release renin > angiotensin II forms > aldosterone released > sodium and water retained > BP rises.

MUSCULOSKELETAL SYSTEM

CONCEPT	KEY FACTS
Osteoblasts vs Osteoclasts	Osteoblasts BUILD new bone (think: Build). Osteoclasts break down bone. Balance maintains bone density.
Joint types by ROM	Ball-and-socket (hip, shoulder) = greatest range of motion. Hinge (knee, elbow) = one plane only. Pivot = rotation.

CONCEPT	KEY FACTS
Tendons vs Ligaments	Tendon connects muscle to bone (T for tied to bone). Ligament connects bone to bone (L for Linking bones).
Muscle types	Skeletal = voluntary, attached to bones. Smooth = involuntary, in organ walls. Cardiac = heart only, involuntary.
Bone marrow	Red marrow produces blood cells (hematopoiesis) in flat bones. Yellow marrow stores fat in long bone shafts.

HESI A2 — CRITICAL THINKING (25 Questions on Exam)

PRIORITY FRAMEWORKS

FRAMEWORK	RULE	EXAMPLE
Maslow's Hierarchy	Physiological needs first, then Safety, then Psychosocial	Chest pain (physiological) takes priority over a patient wanting a chaplain visit (spiritual)
ABCs	Airway first, then Breathing, then Circulation — always in this order	O2 saturation of 87% on room air = apply supplemental oxygen immediately
ADPIE	Assess > Diagnose > Plan > Implement > Evaluate — never skip or reorder	Always assess before implementing. Always evaluate after.
Acute vs Chronic	New or sudden symptoms always take priority over ongoing chronic conditions	New onset confusion takes priority over a patient with chronic back pain rated 4/10
Stable vs Unstable	Unstable patients (changing vitals, new or worsening symptoms) always seen first	Patient with dropping blood pressure is seen before stable post-op patient

THE 8 RIGHTS OF MEDICATION ADMINISTRATION

RIGHT	WHAT TO DO
Right Patient	Verify two identifiers — armband AND ask the patient to state their name
Right Drug	Verify drug name carefully — look-alike and sound-alike drugs are a leading error source
Right Dose	Calculate carefully. Double-check all high-alert medications with a second nurse.
Right Route	PO, IV, IM, SQ, SL, topical — never assume the route; verify the order specifically
Right Time	Scheduled times, PRN parameters, and especially time-sensitive antibiotics (within 1 hour)
Right Documentation	Document AFTER giving — never document in advance or for another nurse
Right Reason	Know WHY the medication is ordered. If unclear, look it up before giving it.
Right Response	Assess the patient after administration. Did it achieve the expected therapeutic effect?

DELEGATION RULES CNAs and NAs CAN do: vital signs on stable patients, ADL assistance, I&O; measurement, non-sterile specimen collection, ambulation assistance, basic personal care. CNAs and NAs CANNOT do: assessment, nursing diagnosis, care planning, medication administration, IV insertion or management, sterile procedures, patient teaching, or evaluation of care outcomes. The nurse retains accountability.

TEAS 7 — READING (52 Scored Questions)

TEXT STRUCTURES — Identify These Instantly

STRUCTURE	SIGNAL WORDS	NURSING EXAMPLE
Cause and Effect	because, therefore, as a result, consequently, thus	Handwashing reduces hospital-acquired infections by 40%
Compare and Contrast	however, similarly, both, on the other hand, whereas, unlike	Hand sanitizer versus soap and water effectiveness
Sequential/Chronological	first, then, next, finally, step 1, step 2, subsequently	Steps for performing hand hygiene procedure
Problem and Solution	problem, issue, solution, resolve, therefore, as a result	Nurse burnout and evidence-based staffing solutions
Description/Definition	for example, specifically, characteristics, defined as, consists of	Describing characteristics of a wound or skin lesion

EVIDENCE HIERARCHY — Strongest to Weakest

LEVEL	TYPE	STRENGTH / NOTES
Level 1	Systematic review or Meta-analysis of RCTs	STRONGEST — aggregates multiple studies, highest quality evidence
Level 2	Randomized Controlled Trial (RCT)	Controlled, randomized — gold standard for individual studies
Level 3	Cohort or Case-control study	Observational — good for rare diseases or long-term outcomes
Level 4	Case study or Case series	Limited generalizability — useful for generating hypotheses
Level 5	Expert opinion or Clinical experience	WEAKEST — subjective, may reflect bias — use when no research exists

KEY READING SKILLS

SKILL	DEFINITION	HOW TO APPLY IT
Main Idea	The central point the passage is making	Look at topic sentences. Ask: what is this mostly about?
Inference	A logical conclusion drawn from evidence not directly stated	Combine text clues with prior knowledge. Must be supported by evidence.
Author Purpose	Why the author wrote the passage (inform, persuade, entertain)	Research articles inform. Opinion pieces persuade. Narratives entertain.
Fact vs Opinion	Fact = verifiable. Opinion = judgment or value statement	Watch for: best, most important, should, I believe, worst.
Context Clues	Surrounding words that help define an unfamiliar term	"The febrile child had a temperature of 104 F" — febrile = having a fever

TEAS 7 — MATH (45 Scored Questions)

NUMBER OPERATIONS — Quick Reference

OPERATION	RULE	EXAMPLE
Fraction addition	Find LCD, convert both fractions, then add numerators	3/8 + 5/12: LCD=24, so 9/24 + 10/24 = 19/24
Fraction multiplication	Multiply numerators together, multiply denominators together, simplify	2/3 x 9/4 = 18/12 = 3/2
Decimal to percent	Multiply decimal by 100	0.625 x 100 = 62.5%
Percent to decimal	Divide percent by 100	37.5 / 100 = 0.375
Percent of a number	Convert to decimal then multiply	35% of 240 = 0.35 x 240 = 84
Percent change	(New minus Old) divided by Old, times 100	Weight 80 to 86 kg: (6/80) x 100 = 7.5% increase
Order of operations	PEMDAS: Parentheses, Exponents, Multiply/Divide left to right, Add/Subtract	(4 squared x 3) minus (2 cubed + 5) = 48 minus 13 = 35
Solving for x	Use inverse operations to isolate the variable on one side	5x minus 3 = 12 gives 5x = 15 gives x = 3
Ratio/Proportion	Set up equivalent fractions and cross-multiply to solve	400mg/10mL = 600mg/x gives x = 15 mL
Absolute value	Distance from zero — always positive	The absolute value of negative 15 plus the absolute value of 8 = 15 + 8 = 23

STATISTICS AND DATA — Know These Terms

TERM	DEFINITION	HOW TO FIND IT
Mean (Average)	Sum of all values divided by the number of values	Add all numbers, then divide by the count
Median	The middle value when all values are placed in order	Order values from least to greatest, then find the middle
Mode	The value that appears most frequently in the data set	Count how often each value appears, pick the most common
Range	The spread of the data from lowest to highest	Maximum value minus minimum value
Probability	The likelihood of a specific event occurring	Favorable outcomes divided by total possible outcomes
p-value	Probability that results occurred by chance alone	p below 0.05 = statistically significant (less than 5% chance of being random)
Correlation	A relationship or association between two variables	Does NOT equal causation. A controlled experiment is needed to prove causation.

MEASUREMENT CONVERSIONS FOR TEAS

CONVERT FROM	CONVERT TO	FORMULA	EXAMPLE
Feet	Inches	Multiply by 12	5 feet 9 inches = (5 x 12) + 9 = 69 inches
Pounds	Kilograms	Divide by 2.2	176 lbs / 2.2 = 80 kg
Kilograms	Grams	Multiply by 1,000	2.5 kg x 1,000 = 2,500 g
Liters	Milliliters	Multiply by 1,000	2.5 L x 1,000 = 2,500 mL
Celsius	Fahrenheit	(C x 9/5) + 32	39.4 C: (39.4 x 1.8) + 32 = 102.9 F
Fahrenheit	Celsius	(F minus 32) x 5/9	103.1 F: (71.1) x 5/9 = 39.5 C
Milliliters	Fluid ounces	Divide by 30	250 mL / 30 = 8.3 fl oz

TEAS 7 — SCIENCE (65 Scored Questions)

DNA, RNA AND PROTEIN SYNTHESIS

CONCEPT	KEY FACTS
DNA base pairing	A pairs with T (adenine-thymine). G pairs with C (guanine-cytosine). Memory: AT the GC meeting.
RNA base pairing	A pairs with U in RNA (uracil replaces thymine). G still pairs with C.
Transcription	DNA is copied into mRNA. Occurs in the NUCLEUS. Think: transcribing the DNA code into a message.
Translation	mRNA is read at RIBOSOMES to build a protein. tRNA brings specific amino acids.
mRNA (messenger)	Carries the genetic code from the nucleus to the ribosome for protein synthesis.
tRNA (transfer)	Carries specific amino acids to the ribosome. Matches codons on mRNA.
rRNA (ribosomal)	Structural component that makes up the ribosome itself.
Protein structure	Primary (amino acid sequence) > Secondary (helix or sheet) > Tertiary (3D fold) > Quaternary (multiple chains)
Mitosis result	Produces 2 IDENTICAL DIPLOID cells with 46 chromosomes each. Used for growth and tissue repair.
Meiosis result	Produces 4 HAPLOID gametes with 23 chromosomes each. Used to create sperm and egg cells.

SCIENTIFIC METHOD AND RESEARCH DESIGN

STEP	DESCRIPTION	KEY VOCABULARY
1. Observation	Notice a phenomenon in the natural world	Qualitative (descriptive) vs quantitative (numerical)
2. Question	Identify what you want to explain or investigate	Must be answerable through systematic investigation
3. Hypothesis	Educated, testable, and falsifiable prediction	If..then..because format. Must be able to be proven wrong.
4. Experiment	Controlled test designed to test the hypothesis	IV = what changes. DV = what is measured. Control group = no treatment.
5. Analysis	Interpret and evaluate the collected data	Mean, median, p-value, statistical significance, correlation
6. Conclusion	Determine whether data supports or refutes the hypothesis	Correlation does NOT prove causation — a key principle
7. Communication	Share findings through peer review and publication	Watch for publication bias and replication failures

Research Design Key Terms: Independent Variable (IV) = what the researcher changes or manipulates Dependent Variable (DV) = what is measured (the outcome) Control group = receives no treatment — provides the baseline for comparison Placebo = inactive treatment given to control group to account for psychological effects Double-blind = neither participants NOR researchers know who received the real treatment

TEAS 7 — ENGLISH & LANGUAGE USAGE (38 Scored Questions)

GRAMMAR RULES MOST TESTED ON TEAS 7

RULE	CORRECT USAGE	ERROR TO AVOID
Subject-verb agreement — each/every	Each nurse IS responsible	Each nurse ARE responsible
Pronoun case after prepositions	Between you and ME	Between you and I
Parallel structure in lists	Assessing, documenting, administering	To assess, documenting, administer
Apostrophe — plural possessive	The patients' room (multiple patients)	The patients room (missing apostrophe)
Double negative	There were no adverse reactions	There weren't no adverse reactions
Dangling modifier	Running down the hall, the NURSE heard the alarm	Running down the hall, the alarm was loud
Active vs passive voice	The nurse administered the medication	The medication was administered by the nurse
Affect (verb) vs Effect (noun)	Stress AFFECTS healing. Stress has an EFFECT on healing.	Stress has an AFFECT on healing.
Semicolon use	The patient was stable; however, monitoring continued.	The patient was stable, however monitoring continued.
Neither/nor — nearest subject agreement	Neither the nurses nor the doctor IS available	Neither the nurses nor the doctor ARE available

HIGH-VALUE VOCABULARY FOR TEAS 7

WORD	MEANING	ANTONYM / RELATED WORD
Benevolent	Kind, charitable, well-meaning toward others	Malevolent = wishing harm to others
Malignant	Cancerous; tending to spread and become worse	Benign = harmless, noncancerous
Ameliorate	To improve or make a bad situation better	Exacerbate = to worsen or aggravate
Viable	Capable of surviving, working, or succeeding	Nonviable = unable to survive
Ubiquitous	Seeming to appear or be found everywhere	Rare, scarce, uncommon
Empirical	Based on direct observation, experience, or experiment	Theoretical = not yet tested
Circumspect	Wary and cautious; considering all circumstances before acting	Reckless, impulsive
Prophylactic	Preventive — given or used before a problem occurs	Therapeutic = treating an existing problem
Contraindicated	Should NOT be used because it may cause harm	Indicated = recommended for use
Iatrogenic	Caused by medical treatment or healthcare providers	Idiopathic = unknown cause

MASTER CHEAT SHEET 1 — Critical Lab Values

CRITICAL LAB VALUES — Know Normal Range AND Critical Thresholds

LAB VALUE	NORMAL RANGE	CRITICAL LOW	CRITICAL HIGH	NURSING SIGNIFICANCE
Sodium (Na+)	135-145 mEq/L	Below 120	Above 160	Neurological symptoms at both extremes. Seizure risk with severe hyponatremia.
Potassium (K+)	3.5-5.0 mEq/L	Below 2.5	Above 6.5	Cardiac arrhythmia risk at both extremes. EKG changes. Notify provider immediately.
Glucose (fasting)	70-100 mg/dL	Below 50	Above 500	Hypoglycemia = give glucose. Hyperglycemia = insulin. DKA risk above 300.
Calcium (Ca++)	8.5-10.5 mg/dL	Below 7	Above 13	Low: tetany, seizures, Chvosteks sign. High: kidney stones, bone pain, confusion.
Hemoglobin	12-17 g/dL	Below 7	Above 20	Below 7 is typical transfusion threshold. Assess for fatigue, pallor, dyspnea.
WBC	5,000-10,000/uL	Below 2,000	Above 30,000	Low = infection risk (neutropenia). High = infection or inflammation or leukemia.
Platelets	150,000-400,000	Below 50,000	Above 1,000,000	Below 50K = bleeding risk. Avoid invasive procedures. Report to provider.
INR (warfarin)	2.0-3.0 (therapeutic)	Below 1.5	Above 4.0	Above 4.0 = bleeding risk. Hold warfarin. Notify provider. May need vitamin K.
BUN	7-20 mg/dL	—	Above 100	Elevated = kidney dysfunction or dehydration. Check creatinine together.
Creatinine	0.6-1.2 mg/dL	—	Above 10	Best single marker of kidney function. Rises when GFR falls.
Arterial pH	7.35-7.45	Below 7.20	Above 7.60	Below 7.35 = acidosis. Above 7.45 = alkalosis. Critical for ABG interpretation.
pO2 (arterial)	80-100 mmHg	Below 60	—	Below 60 = respiratory failure. Requires supplemental oxygen immediately.
Troponin I	Below 0.04 ng/mL	—	Any elevation	Any elevation above normal indicates myocardial injury. Order 12-lead EKG.

MASTER CHEAT SHEET 2 — Vital Signs and Mnemonics

NORMAL VITAL SIGNS BY AGE GROUP

AGE GROUP	HR (bpm)	RR (per min)	SYSTOLIC BP	TEMP (F)
Newborn (0-1 month)	100-160	40-60	60-90	97.7-99.5
Infant (1-12 months)	100-160	25-50	70-100	97.7-99.5
Toddler (1-3 years)	90-150	20-30	80-110	97.7-99.5
School age (6-12 yr)	70-110	18-25	90-120	97.0-99.0
Adolescent (13-18 yr)	60-100	12-20	100-130	97.0-99.0
Adult (18-64 years)	60-100	12-20	Below 120 normal	97.0-99.0
Older adult (65+)	60-100	12-20	Below 130 target	96.0-98.6

MUST-KNOW NURSING MNEMONICS

MNEMONIC	STANDS FOR	USED FOR
ADPIE	Assess, Diagnose, Plan, Implement, Evaluate	The nursing process — always in this order
ABCs	Airway, Breathing, Circulation	Emergency priority order — airway is always first
FAST	Face drooping, Arm weakness, Speech difficulty, Time to call 911	Stroke recognition and response
SAMPLE	Symptoms, Allergies, Medications, Past history, Last meal, Events	Patient history taking during assessment
PQRST	Provokes, Quality, Radiates, Severity, Time	Pain assessment — complete picture of any pain complaint
RACE	Rescue, Alarm, Confine, Extinguish or Evacuate	Fire safety response in hospital settings
PASS	Pull, Aim, Squeeze, Sweep	Correct technique for using a fire extinguisher
OLDCART	Onset, Location, Duration, Character, Aggravating, Relieving, Treatment	Comprehensive symptom assessment tool
SLUDGE	Salivation, Lacrimation, Urination, Defecation, GI distress, Emesis	Signs of cholinergic toxicity or organophosphate poisoning
PMAT	Prophase, Metaphase, Anaphase, Telophase	Stages of mitosis in order — metaphase = middle
FANBOYS	For, And, Nor, But, Or, Yet, So	Coordinating conjunctions for grammar questions
PEMDAS	Parentheses, Exponents, Multiply, Divide, Add, Subtract	Order of operations for all math calculations
AT the GC meeting	Adenine-Thymine, Guanine-Cytosine	DNA base pairing rules for biology and science questions

MASTER CHEAT SHEET 3 — Electrolyte Imbalances

ELECTROLYTE IMBALANCES — Clinical Signs to Know

ELECTROLYTE	IMBALANCE	CLINICAL SIGNS	COMMON CAUSES
Sodium	Hyponatremia — below 135	Confusion, headache, seizures, cerebral edema, nausea, muscle cramps	Excess water intake, SIADH, heart failure, diuretics
Sodium	Hypernatremia — above 145	Extreme thirst, agitation, dry mucous membranes, lethargy, seizures	Dehydration, diabetes insipidus, excessive sodium intake
Potassium	Hypokalemia — below 3.5	Muscle weakness, cramps, decreased bowel sounds, U wave on EKG, arrhythmias	Vomiting, diarrhea, diuretics, inadequate dietary intake
Potassium	Hyperkalemia — above 5.0	Peaked T waves on EKG, bradycardia, fatal arrhythmia, muscle weakness, paresthesias	Renal failure, ACE inhibitors, tissue destruction, excessive supplementation
Calcium	Hypocalcemia — below 8.5	Tetany, positive Chvosteks sign (facial twitch), Trousseaus sign, prolonged QT, seizures	Hypoparathyroidism, vitamin D deficiency, pancreatitis, renal failure
Calcium	Hypercalcemia — above 10.5	Bones (pain), groans (GI distress), moans (psych), stones (kidney), constipation, lethargy	Hyperparathyroidism, cancer, prolonged immobility, excess vitamin D
Magnesium	Hypomagnesemia — below 1.5	Tremors, seizures, cardiac arrhythmias, hypokalemia and hypocalcemia often accompany it	Alcoholism, malnutrition, prolonged diarrhea, diuretics
Magnesium	Hypermagnesemia — above 2.5	Loss of deep tendon reflexes (earliest sign), respiratory depression, cardiac arrest	Renal failure, excessive IV magnesium administration (eclampsia treatment overdose)

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